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Implementation of an Asset Tracking Embedded System using Ultra-Wide Band Technology

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Abstract

In the most complex indoor environments, such as a warehouse, where there is a large volume of equipment and the possibility of losing it is inevitable, this study reveals an embedded system that uses Ultra-Wide Band (UWB) accuracy to deliver pinpoint asset tracking. It is crucial to keep an eye on them. constructing a Real-Time Locating System (RTLS), to use technical terminology. Even though they were effective, conventional navigation systems like GPS, Wi-Fi, and BLE have limitations due to interference and range. UWB effectively overcomes these constraints with its massive 7.5GHz bandwidth and lower interference (- 41.3dBm/1MHz). When it comes to interior navigation, UWB works better since it has a higher connection than Wi-Fi and GPS. Conversely, UWB has no propagation effects like as scattering or reflection and can simply flow through an obstruction. The goal is to create an embedded system that uses an Arduino UNO to gather data from a Deca Wave DW1000, which will serve as a tag and anchor for effective asset navigation utilizing the Time of Flight (ToF) localization method.

Keywords: Time of Flight (ToF); Decawave Shield (DWS1000); Time Difference of Arrival; Arduino UNO; Geometric Dilution of Precision (GDOP); Line of Sight; Non-Line-of-sight; Time of Arrival (ToA); Triangulation; GPS; Wi-Fi.

1. Introduction

Since 1950 tracking devices have always been in existence and the necessity of the devices is growing in various fields. Right from defense to healthcare, every field is utilizing the fruits of efficient tracking devices. The portability of the devices has also been taken into consideration and today navigation systems such as GPS, Zigbee, and Bluetooth are just embedded on an SoC. Though these are useful and powerful in indoor environments [1-4] and for low power

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requirements due to an increase in technological advancements, consumers' expectations to have a high precision and low latency navigation device are ever-growing [5-10]. This is where Ultra-wideband (UWB) comes into the picture. UWB technology is the one that can efficiently meet the index set by users all over the world. UWB is a present-day technology where speed and accuracy are the two most important parameters that decide a product's efficiency. UWB offers a wide range of up to 200m, centimeter range accuracy, milli second precision, and can penetrate through obstacles seamlessly. UWB must be the most preferable navigation system which is the reason why companies like Apple and Samsung have developed Air Tag and Samsung's SmartThings which track devices with spot-on precision and display real-time coordinates on a Graphical User Interface (GUI).

Working on the principle of Time-of-flight, the Deca wave DWS1000 UWB transceiver is an efficient Arduino UNO shield that can detect the presence of a tag which is the transmitter of a UWB pulse. In this proposed work, the time taken by the UWB pulse to travel from the transmitter to the receiver is converted to the distance through which the position of the tag can be obtained. Along with this, the power with which the UWB signal is received is also being calculated.

The related work [11] presents an alternative method and demonstrates the superiority of Time Difference of Arrival (TDoA) over Time of Arrival (ToA). The article [12] describes the design of a low power and low latency UWB tag and anchor design where Real-time 3D Angle of Arrival (AoA) estimation is used for anchor design. The research gap in the ULoc 3D tracking algorithm lies in several areas that can significantly enhance its performance. Difference Time Difference of Arrival (DTDoA), however, had a limit rate set on the number of anchors, which restricted the update rate in DTDoA. Among these are fixing mistakes brought on by ground reflections at specific altitudes.

Additionally, it is possible to integrate navigation technologies like BLE and UWB, which have the benefit of allowing mobile phones to serve as monitoring devices. Nonetheless, the quality of object detection may be compromised by the absence of localization method implementation, and their use would greatly improve accuracy. [13-15]. With a range of up to 200 meters, ultra-wideband offers ten times the coverage of both Bluetooth and Wi-Fi combined. Though this proposed work achieves the desired range, because of the limited processing capacity of Arduino (whose usage is justified in the latter sections), further accuracy and precision can be achieved with a relatively efficient microcontroller. As a result, the tested range for the proposed system should not be too small [16] to avoid underutilizing UWB technology, which is primarily designed to deliver precise and long-range object identification in indoor localities [17].

2. Proposed Methodology

The UWB module used for equipment tracking is Deca wave DWS1000. The DWS1000 module is a powerful UWB Arduino shield device that combines all the essential components for two-way ranging. It has a Deca wave DWM1000 transceiver mounted on it with a pulse repetition frequency working every 2ns. It offers high precision (10 cm) and data rates (up to 6.8 Mbps), making it ideal for various applications, from asset tracking to indoor navigation. Ultra-wideband modules essentially consist of two parts: an anchor that receives the signals and calculates the distance between the tag and receiver, and a tag that acts as a transmitter of UWB pulses. This distance would be translated into the position coordinates of the tag, which would be attached to an asset. To do this, several of the localization techniques offered by the literature are employed. There is a wide variety of localization algorithms that use intersecting spheres or hyperboloids with a constant time difference, such as TDoA, ToA, AoA, Trilateration, and others.

The architecture of these algorithms regulates how UWB modules communicate with one another and determines the asset's coordinates with accuracy. Each algorithm has specific benefits and drawbacks, and how well it works depends on how accurate and effective the receiver identification is. The methodology selected for location determination indicated that more anchors would be used. In this proposed work, using the DWS1000 obtained the received power and two-way ranging of UWB signals. The tests were conducted in different environments and both LOS and NLOS paths. The tests follow a one-dimensional Time of flight algorithm technique as only a single tag and single anchor were employed in testing. The scenarios in which the features of the UWB signal were tested are tabulated in the successive sections. This work implemented the Time Difference in Arrival algorithm for locating an object's coordinates and the Time of Flight (ToF) algorithm for obtaining the accurate range of an object.

2.1. Time Difference of Arrival (TDoA)

UWB method called Time Difference of Arrival (TDoA) compares the arrival times of signals at several receiver nodes to determine the target's location. Because TDoA can withstand multipath interference and NLOS circumstances, it is a good choice for environments where signal transmission can be difficult. For precise location, additional infrastructure like synchronized anchor nodes might be needed. The Deca wave DWM1000 UWB module also uses TDoA to accurately determine a device's position. It works by transmitting UWB signals to a mobile device from several anchor nodes that are in exact places. After receiving these signals, the mobile device calculates the difference in time between their arrivals. Based on these temporal discrepancies, it determines the distance to each anchor node using the speed of light. The module triangulates its position by intersecting these measurements of separation.

$$TDoA = TDoA_{treceiverA} - TDoA_{treceiverB} \tag{1}$$

It compares the time difference of UWB transmitted signals at multiple receiver nodes which are anchors in this case. Among all the algorithms that were used [12] a balanced approach using TDoA is followed where accuracy was not compromised and at the same time, power consumption would also be minimal.

2.2. Time of Flight Methodology (ToF)

ToF is the algorithm employed to calculate the position of the tag. In this technique, the tag watches for synchronization signals from the anchor nodes to ascertain its position. After the tag receives the synchronization signal, it sends back its signal to the anchor nodes. After receiving the signal from the tag, the anchor nodes calculate how long it takes for the signal to reach the anchor. Because UWB technology has such great accuracy, this time measurement, which is based on the time difference between the signal's transmission and reception, would be quite accurate. The ToF algorithm determines the tag's position about the anchor nodes using the known positions of the anchor nodes and the measured time differences between the tag and multiple anchor nodes.

A single tag and single anchor are used in the proposed work to calculate the position of the tag and to obtain the received power. To calculate the tag's position, multi-lateration techniques are frequently used. The tag's location can be converted into absolute coordinates inside the local coordinate system once it has been determined about the anchor nodes. The time it takes for a signal to travel from a transmitter to a receiver is measured by ToA. Conversely, ToF measures the time it takes for a signal to go back and forth between the transmitter and receiver to determine the distance between them. Essentially, ToF determines distance by examining the signal's round-trip time, whereas ToA concentrates on measuring the amount of time it takes for a signal to propagate from one location to another. Time-of-Flight method also helps UWB achieve its basic functionality, which is to follow a moving object with maximum precision. To determine the distance between two devices, the proposed model makes use of the time-of-flight concept, in which the devices exchange messages including timestamp information. Each device's software controls all aspects of communication, including addressing reception issues and delayed transmissions, to precisely calculate the time it takes for signals to travel between devices. To reduce communication failures and guarantee dependable functioning in a variety of situations, an error-handling technique using Frame Check Sequence (FCS) is added. The model also incorporates calibration elements to improve the accuracy of distance predictions by correcting for systematic mistakes or biases in the measuring equipment. The suggested hardware model has undergone extensive testing and validation, and the results show that it is very accurate and reliable.

Among all the algorithms that were used [12], a balanced approach using the Time-of-Flight technique which is explained in the following section where accuracy was not compromised and at the same time power consumption would also be minimal as per the approach taken.

3. Proposed Hardware Methodology and Equations

3.1. Model For Simulation

A key component of UWB technology is the UWB tag, a UWB signal generator that transmits brief, precisely timed bursts of UWB (Ultra-Wideband) signals. These UWB signals are distinguished by their huge bandwidth and short duration, which makes signal time-of-flight measurement more accurate. UWB pulses are released by the tag (or tags) at predetermined, regular intervals. For later localization, these pulses will be detected by stationary wall-mounted anchors using the Time Difference of Arrival (TDoA) approach.

First, every anchor is synchronized. It takes at least four anchors to pinpoint each tag's location. The anchor nearest the corresponding tag is called the master anchor; it oversees all subsequent communication. In this study, the Deca Wave DWM1000 UWB module uses an Arduino UNO microcontroller. UWB pulses would propagate from the tag to corresponding anchors using this interface. An anchor that can be regarded as a master anchor would receive the pulse, and it would then transmit the pulse timing information to the neighboring anchors via communication protocols like SPI or Ethernet. The TDoA algorithm, which would be run on a server a Raspberry Pi or an Arduino UNO, for example - conducts geometric calculations using triangulation algorithms to determine the location of the tag attached to the hardware. This entire idea is depicted in Fig 1.

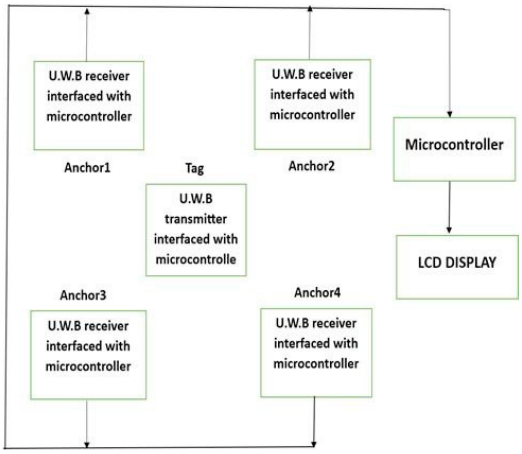

Fig. 1. Block diagram of the proposed hardware methodology

3.2. TDoA Calculation

TDoA calculates the time difference of propagated UWB signals from the transmitter to each receiver/anchor and uses geometric calculation thereby giving the location coordinates of the equipment. The distance between each anchor can be calculated using equations (2 – 6) [18]: -

$$(x - x\_i)^\wedge 2 + (y - y\_i)^\wedge 2 = d\_ijkl^\wedge 2 \quad (2)$$

$$(x - x\_j)^\wedge 2 + (y - y\_j)^\wedge 2 = d\_ijkl^\wedge 2 \quad (3)$$

$$(x - x\_k)^\wedge 2 + (y - y\_k)^\wedge 2 = d\_ijkl^\wedge 2 \quad (4)$$

$$(x - x\_l)^\wedge 2 + (y - y\_l)^\wedge 2 = d\_ijkl^\wedge 2 \quad (5)$$

i,j,k,l are four anchors respectively and d is the distance

$$d\_ijkl = v * t\_ijkl \quad (6)$$

3.3. Model for Obtaining the Range

The Decawave DWS1000 is equipped with a DWM1000 UWB transceiver that produces UWB pulses. In addition to testing message transmission and reception from the module, it also examined the signal quality that a UWB receiver can pick up in various conditions while the tag is moving all the time. In the following sections, the observations are documented. Time-of-Flight method also helps UWB achieve its basic functionality, which is to follow a moving object with maximum precision. To determine the distance between two devices, the model makes use of the time-of-flight concept, in which the devices exchange messages including timestamp information. Each device's software controls all aspects of communication, including addressing reception issues and delayed transmissions, to precisely calculate the time it takes for signals to travel between devices. To reduce communication failures and guarantee dependable functioning in a variety of situations, added error-handling techniques using Frame Check Sequence Technique (FCS). The model also incorporates calibration elements to improve the accuracy of distance predictions by correcting for systematic mistakes or biases in the measuring equipment. The suggested hardware model has undergone extensive testing and validation, and the results show that it is very accurate and reliable. As a result, it may be used for a variety of purposes, including indoor positioning, asset tracking, and localization in wireless sensor networks.

Using the DW1000Ng library which is a pre-defined library in Arduino IDE for ultrawideband (UWB) communication, the DW1000 chip's functions are used to extract information about the received signal's intensity to calculate the received power. This library facilitated the software and register interaction because of which Arduino usage is justified for the objective. Also, because of DWS1000 Arduino shield the application can be scaled by adding multiple anchors and tags. The sender sends data packets constantly after activation and configuration, and the receiver waits for incoming data. Following receipt, the receiver uses the corresponding method to contact the DW1000 chip and obtain data regarding the power received. This function works internally with the registers or variables on the chip that hold information about the intensity of the signal. The received power, which is usually expressed in decibels (dBm), represents the strength of the signal that the receiver received.

When the fixed node transmits, the time at which the signal is sent is recorded as sendTime. The timeout in the transmission is handled at the software level itself. The timestamp at which the moving node receives the signal is the ReceiveTime(). Correspondingly, the time taken to reply by moving node is considered as replyTime(). The final distance of the fixed node from the moving node is determined from the sendTime ReceiveTime() and replyTime() as per equation (7)

$$\text{Result} = \frac{(round1 \times round2 - reply1 \times reply2)}{(round1 + round2 + reply1 + reply2)} \tag{7}$$

where round1 = round1B - round1A.

round1A: This variable stores the timestamp of the first signal transmission by the fixed node. Calling the function that initiates the transmission and returns the timestamp at the beginning of the transmission yields it. round1B: This variable represents the timestamp when the UWB anchor receives the signal transmitted by the tag and sends its response back. Calling the function that awaits an answer from the anchor and delivers the timestamp after the reception yields it.

reply1: The time it takes for the moving node to respond to a signal from the stationary node is represented by this variable. It is obtained by calling the function which extracts the reply time from the received data.

reply2: It is calculated as the time difference between the moment when the anchor transmits its signal (round1B) and the moment when the tag replies.

round1: The time difference between rounds 1B and 1A is computed.

round2: The time it takes for a signal to return from the anchor to the UWB tag and get a response is represented by this variable. Calling the function that gets the signal's round-trip time from the anchor back to the tag yields it.

4. Results and Discussion

4.1. TDoA Results

TDoA computes the time difference between a tag and the corresponding anchor nodes and among the spatially separated receivers themselves. Object detection using TDoA is preliminarily done by using a combination of Time of arrival (ToA) and a correlation mechanism that happens at the processing hub amongst the signals received from receivers. TDoA Calculations can be done by equations (8-10): -

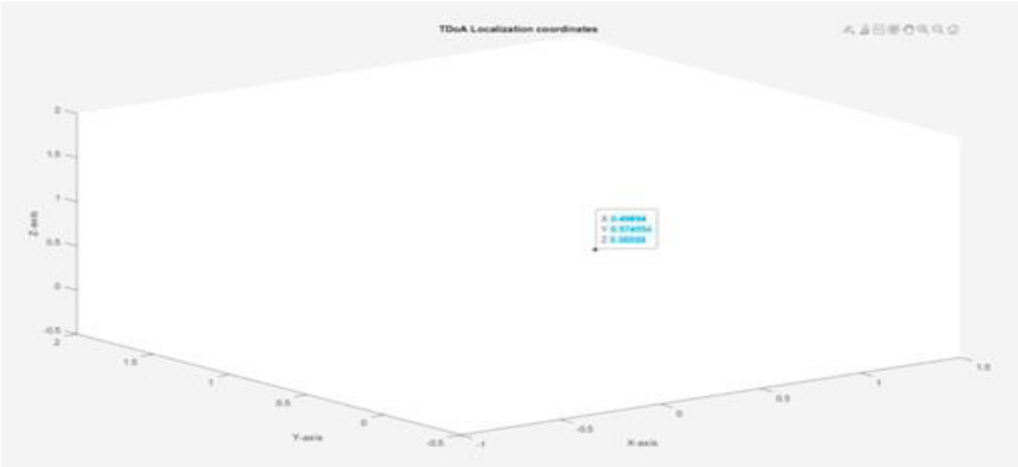

Fig. 2. TDoA Localization coordinated of a tag

TDoA calculations can be done by equations (8 -10): -

$$eqn1 = a == sqrt((x - x2)^2 + (y - y2)^2 + (z - z2)^2) - sqrt((x - x1)^2 + (y - y1)^2 + (z - z1)^2); \tag{8}$$

$$eqn2 = b == sqrt((x - x3)^2 + (y - y3)^2 + (z - z3)^2) - sqrt((x - x1)^2 + (y - y1)^2 + (z - z1)^2); \tag{9}$$

$$eqn3 = c == sqrt((x - x4)^2 + (y - y4)^2 + (z - z4)^2) - sqrt((x - x1)^2 + (y - y1)^2 + (z - z1)^2); \tag{10}$$

where x is a tag and [x1, x2, x3, x4, y1, y2, y3, y4, z1, z2, z3, z4] are the 3D coordinates of 4 geometrically placed anchors. Here subtraction operation is happening with the time difference of a UWB signal to reach from tag to first anchor (Fig 2).

4.1.1. Geometric Dilution of Precision

A minute deviation in the calculations of TDoA can result in a great deal of error in object estimation. This deviation in TDoA measurement is termed as Geometric Dilution of Precision (GDOP) as shown in Fig 3. It should be noted that along the receiver-to-receiver line of sight, accuracy is lowest directly behind the receivers and highest within the envelope created by the network.

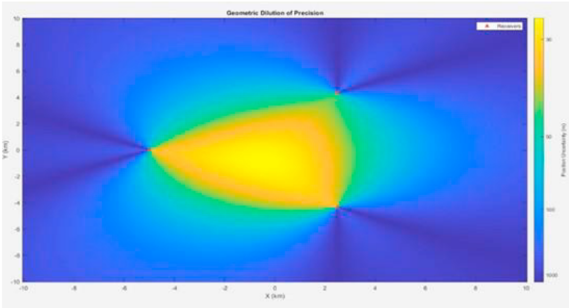

Fig.3. GDOP

4.1.2. Single Tag Tracking

A virtual 2D scenario was created where the receivers which are geometrically positioned constantly monitor the tag which is constantly moving. Initially, the time taken by the tag to transmit signals to anchors would be calculated and the position of the object would be tracked at every step. Filtering of the object is done using Global nearest neighbour (GNN) for better estimation. A single object tracking is depicted as shown in Fig.4.

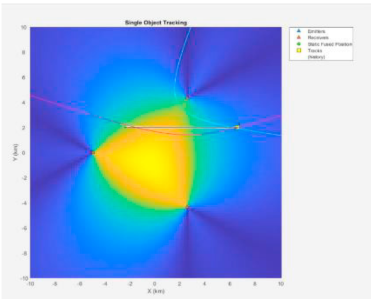

Fig.4. Single Tag Tracking

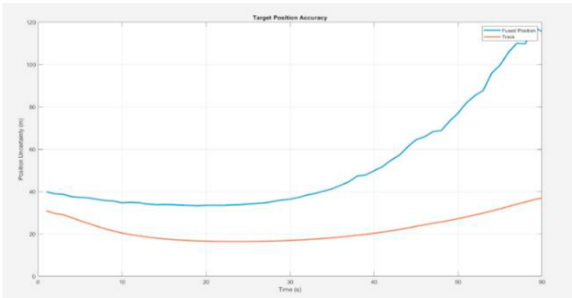

Fig. 5. Estimation of object

The trajectory following the object can be observed from Fig. 5. The object's estimated position is near the point where each receiver pair's hyperbolic curves intersect. Observe that the filtered estimate of the object has less uncertainty at each step than the fused estimate in the zoomed plot below. The geometric dilution of precision causes the estimated position error to increase as the object moves toward the positive X-axis.

The identifier of the tag is shared among multiple anchors/receivers by fusing each object detection individually to obtain their position. This is particularly effective in multiple tags tracking. In single tag tracking, with time the uncertainty of estimating an object is less for the tracker than multiple TDoA measurements fused into one set.

4.1.3. Multiple Object Tracking

When multiple objects are constantly moving and emit multiple UWB pulses, the receiver pairs with an assumption that TDoA calculations among objects are associated and will generate hyperbolas according to the signal identifier. The intersection of hyperbolae would be the object.

Observe how the trajectory is constantly traveling with the object in the range of GDOP (Fig 6). Here, a fusion static algorithm is used to generate the trajectory of multiple objects. As mentioned earlier, this algorithm works effectively when multiple tags are emitting UWB signals. In this manner, the Time Difference of Arrival (TDoA) algorithm can be used to efficiently estimate the target of a tag or multiple tags.

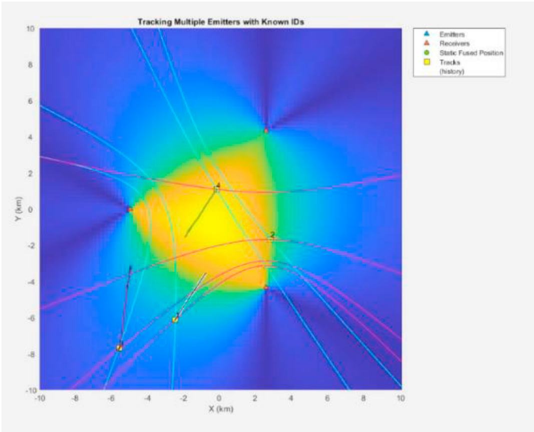

Fig. 6. Multiple tags tracking

4.2. Time of Flight Results

4.2.1. Received Power

There are several ways the UWB pulse gets to the receiver. Based on the transmission and reception of a test message, the strength of the signal is calculated in a scenario where the tag is on the move and the receiver is computing the power calculations according to the range in which the tag is present. Fig.7. contains the signal strength of the UWB pulse when UWB transceivers are exactly placed in the LOS path and at a closer proximity.

The closer proximity can be in the range of a few centimetres to half a meter. When the transmitter and the receiver are separated by about 15 meters, the findings shown in Fig.8 are obtained. Also, created a scenario where the fixed and moving nodes are met with obstacles in an NLOS path where the UWB pulses are penetrated right through them, and efficient readings are obtained as in Fig. 9. Even with a slight shift in the tag's location, the readings are near perfect as they are redirected and updated every second.

4.2.2. Distance of the tag

To obtain pinpoint accuracy, which is the key component of tracking an indoor asset, antenna delay calibration was performed in the proposed work to find out the distance between the tag and anchor. In the case of the DW1000, antenna delay calibration is the process of making up for the time delay that the signal experiences while passing through the antenna system. This can be interpreted through the received power signal strength of the UWB signal. In ultrawideband (UWB) communication systems, the precise determination of the time of flight (ToF) of signals is contingent upon this delay. The values of Fig. 9, which are in millimetre units, give the distance between the tag, which is the actual object/asset to be tracked, and an anchor. Additionally, have set up a situation in which both stationary and mobile nodes encounter obstructions in a non-line-of-sight path. The UWB pulses can pass past these barriers, yielding an accurate distance of the tag, as shown in Fig. 10. With the restricted capabilities of Arduino UNO, up to 13m were able to achieve a near-perfect position of the tag.

4.3. Proof of Accuracy

To prove that UWB can give the position of the tag accurately, a scenario where the position of the tag is known, and the anchor is kept at a fixed distance is created. Fig. 10 depicts this scenario.

```
RX power is [dBm] ... -55.85
Signal quality is ... 84.57
Received message ... #75
Data is ... Hello DW1000Ng, it's #144□
RX power is [dBm] ... -57.10
Signal quality is ... 129.17
Received message ... #76
Data is ... Hello DW1000Ng, it's #146□
RX power is [dBm] ... -56.46
Signal quality is ... 142.53
```

Fig. 7. Received power when the tag is in closer proximity

```
RX power is [dBm] ... -99.53
Signal quality is ... 54.88
Received message ... #58
Data is ... Hello DW1000Ng, it's #151□
RX power is [dBm] ... -100.37
Signal quality is ... 57.67
Received message ... #59
Data is ... Hello DW1000Ng, it's #153□
RX power is [dBm] ... -98.06
Signal quality is ... 50.63
```

Fig. 8. Received power when the tag is at a distance

```
Distance between Tag and Anchor is (in mm) : 790.86
Distance between Tag and Anchor is (in mm) : 810.05
Distance between Tag and Anchor is (in mm) : 799.39
Distance between Tag and Anchor is (in mm) : 1579.92
Distance between Tag and Anchor is (in mm) : 9867.29
Distance between Tag and Anchor is (in mm) : 9856.63
Distance between Tag and Anchor is (in mm) : 9858.76
Distance between Tag and Anchor is (in mm) : 10048.56
Distance between Tag and Anchor is (in mm) : 11643.76
Distance between Tag and Anchor is (in mm) : 13646.29
```

Fig. 9. Distance between tag and anchor

Fig. 10. The Tag is kept at a fixed distance from the anchor.

Possible to locate the tag almost precisely in the instance shown with an error of less than 5 cm, which can be considered negligible, and obtain the distance of the tag. This application may be very useful when an asset must be monitored in an interior setting with the highest level of precision and position as shown in Fig. 11. In this way, by conducting extensive field tests, antenna delay calibration, and using ToF algorithm accuracy and precision of the object can be obtained.

```
Distance between Tag and Anchor is (in mm) : 279.03
Distance between Tag and Anchor is (in mm) : 315.28
Distance between Tag and Anchor is (in mm) : 321.68
Distance between Tag and Anchor is (in mm) : 317.41
Distance between Tag and Anchor is (in mm) : 308.88
Distance between Tag and Anchor is (in mm) : 304.62
```

Fig. 11. Readings for the scenario in Fig. 10

5. Conclusion And Future Work

It was possible to determine the tag's distance in the present study with precision down to the millisecond level and accuracy within a centimeter range. The range was limited to 15 meters due to the ATMEGA processor's functional limitations. It aimed to test the application in more complex and varied circumstances utilizing more efficient processors like the ARM Cortex M series as the end user would be able to get the object's position in real time and with a minimal delay.

References

[1] S. Li, Z. Wang, L. Shi, and X. Zhang, "Experimental in- door tracking testbed based on Visible Light Communication," 2019 26th IEEE International Conference on Electronics, Circuits and Systems (ICECS), Genoa, Italy, 2019, pp. 129-130, doi: 10.1109/ICECS46596.2019.8965061.

[2] M. Contigiani, R. Pietrini, A. Mancini, and P. Zingaretti, "Implementa- tion of a tracking system based on UWB technology in a retail environ- ment," 2016 12th IEEE/ASME International Conference on Mechatronic and Embedded Systems and Applications (MESA), Auckland, New Zealand, 2016, pp. 1-6, doi: 10.1109/MESA.2016.7587123.

[3] A. K. Sansyah, S. O. Ngesthi, B. S. P. P. Ardany, and G. Dewantoro, "Wi- Fi Enabled Asset Surveillance System using Internet of Things," 2021 2nd International Conference on Innovative and Creative Information Technology (ICITech), Salatiga, Indonesia, 2021, pp. 206-211, doi: 10.1109/ICITech50181.2021.9590129.

[4] F. Salinas Go´mez, N. E. Sua´rez Escobar, and J. Va´zquez, "RFID + Wi-Fi system to control the location of biomedical equipment within hospital areas and linked to an intelligent inventory," Health Technol., vol. 10, pp. 479–483, 2020, doi: 10.1007/s12553-019-00393-6.

[5] Q. Tian, K. I. K. Wang, and Z. Salcic, "A Low-Cost INS and UWB Fusion Pedestrian Tracking System," in IEEE Sensors Journal, vol. 19, no. 10, pp. 3733-3740, May 15, 2019, doi: 10.1109/JSEN.2019.2894714.

[6] J. Cho´liz, A. Herna´ndez-Solana, and A. Valdovinos, "Evaluation of al- gorithms for UWB indoor tracking," 2011 8th Workshop on Positioning, Navigation and Communication, Dresden, Germany, 2011, pp. 143-148, doi: 10.1109/WPNC.2011.5961031.

[7] W. Rhee, N. Xu, B. Zhou, and Z. Wang, "Low power, non-invasive UWB systems for WBAN and biomedical applications," 2010 International Conference on Information and Communication Technology Conver- gence (ICTC), Jeju, 2010, pp. 35-40, doi: 10.1109/ICTC.2010.5674731.

[8] R. Kaur, V. Defrancesco, and J. Enderle, "Design, development and evaluation of an asset tracking system," Proceedings of the IEEE 31st Annual Northeast Bioengineering Conference, 2005, Hoboken, NJ, USA, 2005, pp. 110-111, doi: 10.1109/NEBC.2005.1431949.

[9] H. C. Keong and M. R. Yuce, "Low data rate ultra-wideband ECG monitoring system," 2008 30th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, Vancouver, BC, Canada, 2008, pp. 3413-3416, doi: 10.1109/IEMBS.2008.4649939.

[10] M. Ridolfi, S. Velde, H. Steendam, and E. De Poorter, "Analysis of the Scalability of UWB Indoor Localization Solutions for High User Densities," Sensors, vol. 18, no. 6, 2018, doi: 10.3390/s18061875.

[11] L. Santoro, M. Nardello, D. Brunelli and D. Fontanelli," UWB-Based In- door Positioning System With Infinite Scalability," in IEEE Transactions on Instrumentation and Measurement, vol. 72, pp. 1-11, 2023, Art no. 1005711, doi: 10.1109/TIM.2023.3282299. J. Clerk Maxwell, A Treatise on Electricity and Magnetism, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.

[12] Minghui Zhao, Tyler Chang, Aditya Arun, Roshan Ayyalasomayajula, Chi Zhang, and Dinesh Bharadia. 2021. ULoc: Low-Power, Scalable and cm-Accurate UWB-Tag Localization and Tracking for Indoor Ap- plications. Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. 5, 3, Article 140 (Sept 2021), 31 pages. https://doi.org/10.1145/3478124K. Elissa, "Title of paper if known," unpublished.

[13] KAR˙IMPOUR, Noushin & Dagdeviren, Zuleyha & AKRAM, Vahid & DAG˘ DEV˙IREN, Orhan. (2021). A Cloud-based Asset Tracking System for Hospitals Using Ultra-wideband Localization. European Journal of Science and Technology. 10.31590/ejosat.960454.Y. Yorozu, M. Hirano,

[14] S. Shyam, S. Juliet, and K. Ezra, "Tracking and Monitoring of Medical Equipments using UWB for Smart Healthcare," 2022 6th International Conference on Computing Methodologies and Commu- nication (ICCMC), Erode, India, 2022, pp. 631-637, doi: 10.1109/IC- CMC53470.2022.9753719.

[15] A. K. Sansyah, S. O. Ngesthi, B. S. P. P. Ardany, and G. Dewantoro, "Wi- Fi Enabled Asset Surveillance System using Internet of Things," 2021 2nd International Conference on Innovative and Creative Information Technology (ICITech), Salatiga, Indonesia, 2021, pp. 206-211, doi: 10.1109/ICITech50181.2021.9590129.

[16] P. Grasso, M. Innocente, J. J. Tai, O. Haas, and A. M. Dizqah, "Analysis and Accuracy Improvement of UWB-TDoA-Based Indoor Positioning System," Sensors, vol. 22, no. 9136, 2022, doi: 10.3390/s22239136.

[17] K. Naheem and M. S. Kim, "A Robust Indoor Pedestrian Lo- calization Approach Against Human Body Shadowing for UWB- Enabled Smartphones," in IEEE Transactions on Instrumentation and Measurement, vol. 73, pp. 1-13, 2024, Art no. 9505013, doi: 10.1109/TIM.2024.3368497.

[18] M. J. Kuhn, C. Zhang, S. Lin, M. R. Mahfouz, and A. E. Fathy, "A system level design approach to UWB localization," 2009 IEEE MTT-S International Microwave Symposium Digest, Boston, MA, USA, 2009, pp. 1409-1412, doi: 10.1109/MWSYM.2009.5165970.